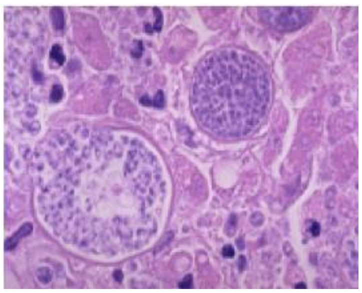
IS-NPMD- Dermatology
Skin and wound infections-Practice Questions with Answer key
Faculty: Dr. Pai

The next two questions are associated with this case.

A 50-year-old farmer from Arizona was referred to the hospital because of malaise, weakness, weight loss and severe joint pains. Examination of the limbs revealed multiple nodular lesions on his shin. A biopsy of the lesion was carried out and a microscopic picture of the biopsy sample is attached.

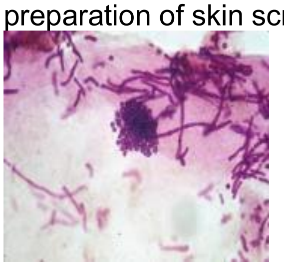


1. What is the best description of the pathogen that caused this infection?
   a. A dimorphic fungus which forms spherules. Coccidioides
   b. A bacterium that is transmitted through ticks.
   c. A dimorphic fungus that produces broad based buds.
   d. A parasite which is acquired through anopheles mosquito
   e. A yeast-like fungus which produces chlamydospores

2. What most likely led to his infection?
   a. Wound contamination with sea water
   b. Cat scratch or bite Bartonella
   c. Tick bite
   d. Inhalation of arthroconidia
   e. Rose thorn prick Sporothrix schenckii

The next two questions are associated with this case.

A 34-year-old man presents to his physician with hypopigmented lesions on his chest and back. On examination the lesions are painless, superficial and dry. A microscopic picture of a KOH preparation of skin scrapings is shown in the attached picture.



3. Which of the following is the most likely etiologic agent?
   a. *Candida albicans*
   b. *Malassezia furfur*
   c. *Microsporum canis*
   d. *Sporothrix schenckii*
   e. *Trichophyton rubrum*

4. Which of the following best describes the characteristics of the pathogen?
   a. Yeast cells and pseudohyphae; germ tube test positive
   b. Macroconidia and microconidia with hyphae; zoophilic
   c. Spaghetti and meatball appearance; grows well in SDA with olive oil
   d. Many microconidia and pencil shaped macroconidia with hyphae; red pigment
   e. Cigar shaped yeasts; dimorphic fungus

The next three questions are associated with this case.

A 23- year- old breast-feeding mother presents with pain, swelling and increased warmth around her left breast. On examination, skin overlying the breast is red, edematous, tender, and hot; Gram stain of the aspirated pus shows gram positive cocci in clusters.

s. aureus

5. Which of the following is a characteristic of the organism seen in the pus?
   a. Thick peptidoglycan layer and teichoic acid
   b. Thin peptidoglycan layer and lipopolysaccharide
   c. Outer membrane made up of lipids
   d. Thick peptidoglycan and endotoxin
   e. Outer lipid membrane and lipoteichoic acid

6. What virulence factors of the most likely pathogen led to her breast abscess?
   a. Hyaluronidase and Streptolysin
   b. Protein A and Coagulase
   c. Exotoxin A and pili
   d. Capsule and tripartite toxin anthrax
   e. Teichoic acid and exfoliatin

Case contd. In the above case, the baby that was breast fed by this mother developed some scaly lesions around his mouth and extensive peeling of skin was seen on his limbs.

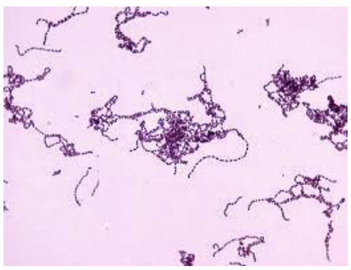
7. Which of the following best explains the mode of action of the virulence factor that led

to the baby's condition?

a. The toxin splits intercellular bridges in the Stratum granulosum of epidermis.
b. The toxin causes nonspecific activation of T cells
c. The enzyme helps in extensive spreading along the connective tissues
d. The enzyme destroys red blood cells and white blood cells
e. The protein helps evade phagocytosis

The next three questions are associated with this case.

A 15-year-old male presents with complaints of pain and swelling in the right leg. He developed a small abrasion when he slipped and grazed the limb while rock climbing. This soon spread along his leg. On examination, the right leg is red and swollen, and is warm and tender to touch. The erythematous lesion does not have any clear demarcation. Vitals-Temperature:101 degree F (38.3 degree C), Pulse 135/min, BP 130/80. A gram stain of the wound aspirate is attached and the pathogen is isolated on aerobic incubation.



8. Which of the following best describes the likely etiologic agent?

a. Streptococcus mutans
b. Staphylococcus aureus
c. Streptococcus pneumoniae
d. Staphylococcus epidermidis
e. Streptococcus pyogenes

9. What is most likely responsible for the spreading nature of this infection?

a. Phospholipase C
b. Hyaluronidase
c. Coagulase
d. Exotoxin A
e. Protective antigen

10. Which of the following complications are known to be associated with the most likely pathogen?

a. TSS and necrotizing fasciitis
b. TSS and scalded skin syndrome
c. Sepsis and meningitis
d. Cellulitis and gastroenteritis
e. Sinus formation with sulfur granules

11. Case- A 44-year-old landscaper presents with the complaint of a non-healing, open sore on the back of his hand and some bumps along his arm. Physical examination reveals an ulcerative lesion on his right hand and several nodular lesions on his right arm along the lymphatic chain. Microscopic examination of the tissue would most likely demonstrate which of the following?

a. elongated yeast
b. gram-positive bacilli
c. gram-positive cocci
d. hyphae with clusters of conidia
e. sulfur granules

The next two questions are associated with this case

A 32-year-old butcher, from Zambia (Africa), visits the local physician with a skin lesion and severe swelling of his right arm. He notes that a few days ago the lesion was a small painful pimple that rapidly progressed into a large ulcer.
O/E, the lesion is ulcerative, 10 mm in diameter and has a black necrotic appearance (see image). The surrounding tissue is severely edematous and there is painful lymphadenopathy.

anthrax

12. Which of the following best describes the causative agent?
a. Gram positive bacilli in chains forming protective antigen with two other factors.
b. Gram positive cocci in chains forming hyaluronidase
c. Gram positive cocci in clusters forming Panton Valentine leucocidin
d. Gram positive filamentous bacilli that are partially acid fast
e. Gram negative bacilli that forms exotoxin A
f. Cigar shaped yeasts that are dimorphic

13. Which of the following best describes the growth characteristic of this pathogen?
a. Beta hemolysis on blood agar
b. Medusa head colonies on blood agar
c. Yeast at 37deg C and mold at 22 deg C
d. Green colonies on TCBS
e. Yellow colonies on Mannitol Salt agar

14. A 27-year-old soldier who has returned from Afghanistan, presents with a very painful erosive ulcer near his pinna. The infection followed bite by sandflies which is commonly found in the region where he was posted. What is the most likely microscopic picture of a punch biopsy taken from the edge of the lesion?

a. Yeast like budding cells of *Candida albicans*
b. Amastigote forms of *Leishmania tropica*
c. Sclerotic cells of dematiaceous molds
d. Arthroconidia of *Coccidioides immitis*
e. Trypomastigote forms of *Trypanosoma cruzi*

15. Which of the following drugs is indicated in the above case?
a. Topical azole
b. Potassium iodide
c. TMP-SMZ
d. Doxycycline
e. Miltefosine

16. A 37-year-old woman visits the dermatologist with symptoms of progressive pruritus over her left arm. She does not have any other major symptoms and is not on any medications. Physical examination reveals an erythematous scaly lesion with a reddish border. Skin scrapings treated with 10% KOH reveals the presence of numerous hyphal fragments. Which of the following is most likely the causal organism?

a. *Trichophyton mentagrophytes.*
b. *Malassezia furfur.*
c. *Sporothrix schenckii*
d. *Coccidioidomyces immitis.*
e. *Candida albicans.*

The next two questions are associated with this case.(You may select more than one option, if applicable)
Following a gun- shot injury, a soldier develops a rapidly progressive massive necrotic ulcerative lesion with extensive swelling and crepitus around the wound.

17. Which of the following will most likely be seen if a wound swab is subjected to grams stain?
a. Acid fast bacilli
b. ~~Gram positive~~ filamentous bacilli
c. Gram positive plump rods
d. Gram negative capsulated bacilli
e. Gram negative cocci

18. Which of the following will be useful to confirm the virulence of the pathogen?
a. Sensitivity to bacitracin
b. Nagler's reaction
c. Catalase test
d. Anti-DNAse B test
e. Detection of mycolic acid

19. What virulence factor is formed by the most likely pathogen?

a. Urease
b. Exotoxin A
c. Phospholipase C
d. Alpha toxin
e. Lecithinase

20. Associate the following conditions with the most likely pathogens listed below. You may choose a pathogen more than once if applicable.

1. Hot Tub folliculitis P. aeruginosa
2. Ecthyma gangrenosum seen in immuno-compromised hosts
3. Extensive burns infections with pus with fruity odor P. aeruginosa
4. Exposure to sea water and cellulitis virbio vulfinicu
5. Injury while handling seals/ whales/ fish followed by violaceous lesions Erysipelothrix rhusiopathiae
6. Skin lesions with loss of sensations and extensive nodular lesions, poor CMI
7. Anaesthetic skin lesions with thickened nerves, good CMI
8. Thorn prick with lympho-cutaneous nodules which can ulcerate sporothrix
9. Cat bite and gram-negative coccobacilli bartonella
10. Cat bite and organisms seen only with silver stain and not grams stain pasturella
11. Human bite, bleach-like odor eikenella
12. Exposure to sandy soil- inhalation of arthroconidia coccidiodes
13. Contact exposure to sandy soil potentially contaminated with larvae
14. Mosquito bite followed by lymphatic involvement and enlarged scrotum/ limbs
15. Sand fly bite followed by nodulo-ulcerative lesion leishmania
16. Deep seated abscesses, foul smelling
17. Fish tank granuloma M. marinum
18. Spreading lesions with extensive and rapidly progressive necrosis of skin and soft tissues and gram-positive cocci in chains
19. Extensive tissue damage with crepitations due to gas trapped in tissues, GPB
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22. Body aches, arthritis, nodular skin lesions, California and Arizona coccidiodes
23. Ulcerated verrucous granulomas, Ohio- Mississippi region, broad based buds
24. Overcrowded conditions, pruritic skin lesions with tracks
25. Painful papule with emerging worm

a. Phialophora
b. Pasteurella multocida
c. Bartonella henselae
d. Strep pyogenes
e. Pseudomonas aeruginosa
f. Vibrio vulnificus
g. Erysipelothrix rhusiopathiae
h. Lepromatous leprosy-M.leprae
i. Tuberculoid leprosy-M.leprae
j. Nocardia spp

k. Sporothrix schenckii
l. Blastomyces dermatitidis
m. Coccidioides immitis
n. Wuchereria/ Brugiya
o. Dracunculus medinensis
p. Eikenella corrodens
q. Mycobacterium marinum(Atypical mycobacteria)
r. Non sporing anaerobes
s. Sarcoptes scabiei
t. Cutaneous Larva Migrans
u. Leishmania tropica
v. Clostridium perfringens

Scroll down for answer key

Answer key:

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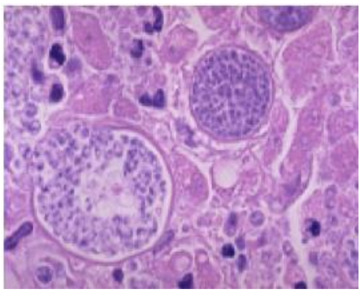

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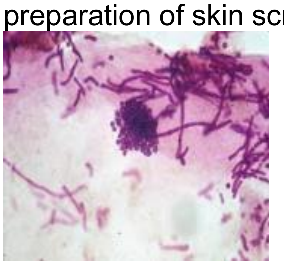

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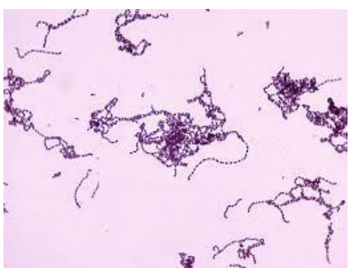

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d. *Coccidioidomyces immitis.*
e. *Candida albicans.*

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20.

1	a
2	d
3	b
4	c
5	a
6	b
7	a
8	e
9	b
10	a
11	a
12	a
13	b
14	b
15	e
16	a
17	c
18	b
19	c/d/e-all are names for the same toxin
20	See matched list given below

20. Associate the following clinical presentations with the most likely pathogens listed below. You may choose a pathogen more than once if applicable.

1. Hot Tub folliculitis- *Pseudomonas aeruginosa*
2. Ecthyma gangrenosum seen in immuno-compromised hosts- *Pseudomonas aeruginosa*
3. Extensive burns infections with pus with fruity odor- *Pseudomonas aeruginosa*
4. Exposure to sea water and cellulitis- *Vibrio vulnificus*

5. Injury while handling seals/ whales/ fish followed by violaceous lesions- *Erysipelothrix rhusiopathiae*
6. Skin lesions with loss of sensations and extensive nodular lesions, poor CMI- *Lepromatous leprosy-M.leprae*
7. Anaesthetic skin lesions with thickened nerves, good CMI- *Tuberculoid leprosy-M.leprae*
8. Thorn prick with lympho-cutaneous nodules which can ulcerate- *Sporothrix schenckii*
9. Cat bite and gram negative coccobacilli- *Pasteurella multocida*
10. Cat bite and organisms seen only with silver stain and not grams stain- *Bartonella henselae.*
11. Human bite, bleach-like odor- *Eikenella corrodens*
12. Exposure to sandy soil- inhalation of arthroconidia- *Coccidioides immitis*
13. Contact exposure to sandy soil potentially contaminated with larvae- *Cutaneous Larva migrans*
14. Mosquito bite followed by lymphatic involvement and enlarged scrotum/ limbs- *Wuchereria/ Brugiya*
15. Sand fly bite followed by nodulo-ulcerative lesion- *Leishmania tropica*
16. Deep seated abscesses, foul smelling- *Non sporing anaerobes*
17. Fish tank granuloma- *Mycobacterium marinum*
18. Spreading lesions with extensive and rapidly progressive necrosis of skin and soft tissues and gram positive cocci in chains- *Strep pyogenes*
19. Extensive tissue damage with crepitations due to gas trapped in tissues, GPB- *Clostridium perfringens*
20. Subcutaneous swelling with multiple sinus tracts, Acid fast bacilli- *Nocardia spp*.
21. Subcutaneous tissue lesions, sclerotic bodies seen- *Phialophora*
22. Body aches, arthritis, nodular skin lesions, California and Arizona- *Coccidioides immitis*
23. Ulcerated verrucous granulomas, Ohio- Mississippi region, broad based buds *Blastomyces dermatitidis*
24. Overcrowded conditions, pruritic skin lesions with tracks- *Sarcoptes scabei*
25. Painful papule with emerging worm- *Dracunculus medinensis*

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